MCS5202 Control Engineering III

State space description of linear systems. Concepts of controllability and observability. Canonical realization of systems having specified transfer functions. Stability in the sense of Lyapunov. State feedback, modal control, pole assignment and the optimal quadratic regulator. Full-order state observers. Multi-lateral systems. Introduction to sampled-data systems. Digital compensation. Introduction to microprocessor-based Control.

Introduction

One of the most important tasks in the analysis and design of control systems is mathematical modeling of the system in which physical systems are represented by Dynamic equations in form of differential equations. The differential equations describing the dynamic performance of a physical system are obtained by utilizing the physical laws of the process (such as Newton's Laws, Kirchhoff's laws, Ohm's law etc.).

There are two approaches for analysis and design of control systems namely:

- Classical (frequency domain) approach
- Modern (time domain) approach

In classical approach the systems differential equations are converted in to transfer function.

Advantage of classical approach

- Rapidly provide stability and transient response information

Disadvantage of classical approach

- Restricted to single input single output (SISO) systems only
- Can only be applied to linear time invariant (LTI) systems

In modern approach the systems differential equations are converted in to a set of first-order differential equations, which may be combined into a first-order vector-matrix differential equation.

Modern approach is a unified approach for modelling, analysing and designing of a wide range of control systems.

Advantage of modern approach

- Can be applied to nonlinear systems
- Applicable to time varying systems
- Easily tackled by the availability of advanced digital computer.
- Applicable to multi input multi output (MIMO) systems

State Space representation

A system is represented in state space using two state equations as follows:

$$\left.\begin{aligned}\underline{\dot{x}}(t) &= A\underline{x}(t) + B\underline{u}(t) && \rightarrow \text{State Equation}\\ \underline{y}(t) &= C\underline{x}(t) + D\underline{u}(t) && \rightarrow \text{Output Equation}\end{aligned}\right\} \text{State Space Equations}$$

$\underline{x}(t)$: State vector

$\underline{\dot{x}}(t)$: Derivative of the state vector

$\underline{u}(t)$: Input vector

$\underline{y}(t)$: Output vector

A : System Matrix

B : Input Matrix

C : Output Matrix

D : Feedforward Matrix

The dimensions of the vectors and matrices are illustrated below:

$$\begin{bmatrix} \dot{x}_1(t) \\ \dot{x}_2(t) \\ \vdots \\ \dot{x}_n(t) \end{bmatrix}_{n\times 1} = [A]_{n\times n} \begin{bmatrix} x_1(t) \\ x_2(t) \\ \vdots \\ x_n(t) \end{bmatrix}_{n\times 1} + [B]_{n\times m} \begin{bmatrix} u_1(t) \\ u_2(t) \\ \vdots \\ u_m(t) \end{bmatrix}_{m\times 1} \quad \text{State Equation}$$

$$\begin{bmatrix} y_1(t) \\ y_2(t) \\ \vdots \\ y_p(t) \end{bmatrix}_{p\times 1} = [C]_{p\times n} \begin{bmatrix} x_1(t) \\ x_2(t) \\ \vdots \\ x_n(t) \end{bmatrix}_{n\times 1} + [D]_{p\times m} \begin{bmatrix} u_1(t) \\ u_2(t) \\ \vdots \\ u_m(t) \end{bmatrix}_{m\times 1} \quad \text{Output Equation}$$

State Space representation of nth order systems of linear differential equation

The following examples illustrate how to represent a system in state space from its differential equation:

Example 1

A system is described by the differential equation

$$\frac{d^3y}{dt^3} + 5\frac{d^2y}{dt^2} + 9\frac{dy}{dt} + 10y = 6u(t)$$

Where y is the output and u is the input of the system. Obtain the state space representation of the system.

Solution

Step 1 The differential equation can be rewritten as:

$$\dddot{y} = 6u(t) - 5\ddot{y} - 9\dot{y} - 10y \qquad (1)$$

Step 2 The state variables can be chosen as:

$$\begin{aligned} x_1 &= y \\ x_2 &= \dot{y} \\ x_3 &= \ddot{y} \end{aligned} \qquad (2)$$

Step 3 The derivatives of the state variables can be obtained as:

$$\begin{aligned} \dot{x}_1 &= \dot{y} \\ \dot{x}_2 &= \ddot{y} \\ \dot{x}_3 &= \dddot{y} \end{aligned} \qquad (3)$$

Step 4 Substituting (2) and (1) in (3) and rearrange gives:

$$\begin{aligned} \dot{x}_1 &= x_2 \\ \dot{x}_2 &= x_3 \\ \dot{x}_3 &= 6u(t) - 5\ddot{y} - 9\dot{y} - 10y \\ &= -10x_1 - 9x_2 - 5x_3 + 6u(t) \end{aligned} \qquad (4)$$

From (2) the output equation is:

$$y = x_1 \tag{5}$$

Step 5 The state equations can be obtained by expressing equations (4) and (5) in matrix form as:

$$\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \\ \dot{x}_3 \end{bmatrix} = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ -10 & -9 & -5 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} + \begin{bmatrix} 0 \\ 0 \\ 6 \end{bmatrix} u$$

$$y = \begin{bmatrix} 1 & 0 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$$

$$A = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ -10 & -9 & -5 \end{bmatrix} \qquad B = \begin{bmatrix} 0 \\ 0 \\ 6 \end{bmatrix} \qquad C = \begin{bmatrix} 1 & 0 & 0 \end{bmatrix} \qquad D = 0$$

Example 2

A single input single output system has the transfer function

$$\frac{Y(s)}{U(s)} = \frac{1}{s^3 + 10s^2 + 29s + 18}$$

Obtain the state space representation of the system.

Solution

The corresponding differential equation can be obtained as:

$$\dddot{y} + 10\ddot{y} + 29\dot{y} + 18y = u$$

Step 1 The differential equation can be rewritten as:

$$\dddot{y} = u - 10\ddot{y} - 29\dot{y} - 18y \tag{1}$$

Step 2 The state variables can be chosen as:

$$\begin{aligned} x_1 &= y \\ x_2 &= \dot{y} \\ x_3 &= \ddot{y} \end{aligned} \tag{2}$$

Step 3 The derivatives of the state variables can be obtained as:

$$\dot{x}_1 = \dot{y}$$

$$\dot{x}_2 = \ddot{y} \qquad (3)$$

$$\dot{x}_3 = \dddot{y}$$

Step 4 Substituting (2) and (1) in (3) and rearrange gives:

$$\dot{x}_1 = x_2$$

$$\dot{x}_2 = x_3 \qquad (4)$$

$$\dot{x}_3 = u - 10\ddot{y} - 29\dot{y} - 18y$$

$$= -18x_1 - 29x_2 - 10x_3 + u$$

From (2) the output equation is:

$$y = x_1 \qquad (5)$$

Step 5 The state equations can be obtained by expressing equations (4) and (5) in matrix form as:

$$\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \\ \dot{x}_3 \end{bmatrix} = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ -18 & -29 & -10 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} + \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} u$$

$$y = \begin{bmatrix} 1 & 0 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$$

Example 3

Consider a system described by the following pair of differential equations

$$J\frac{d^2\theta}{dt^2} + f\frac{d\theta}{dt} - k_m I_a = 0$$

$$k_m\frac{d\theta}{dt} + R_a I_a + L_a\frac{dI_a}{dt} = V(t)$$

Obtain the state space representation of the system. Assume the output of the system to be θ

Solution

The differential equation can be rewritten as:

$$\ddot{\theta} = \frac{k_m}{J} I_a - \frac{f}{J} \dot{\theta}$$

$$\dot{I_a} = \frac{1}{L_a} V - \frac{R_a}{L_a} I_a - \frac{k_m}{L_a} \dot{\theta} \qquad (1)$$

The state variables can be chosen as:

$$x_1 = \theta$$

$$x_2 = \dot{\theta} \qquad (2)$$

$$x_3 = I_a$$

The derivatives of the state variables can be obtained as:

$$\dot{x}_1 = \dot{\theta}$$

$$\dot{x}_2 = \ddot{\theta} \qquad (3)$$

$$\dot{x}_3 = \dot{I_a}$$

Substituting (2) and (1) in (3) and rearrange gives:

$$\dot{x}_1 = x_2$$

$$\dot{x}_2 = \frac{k_m}{J} I_a - \frac{f}{J} \dot{\theta}$$

$$= \frac{k_m}{J} x_3 - \frac{f}{J} x_2$$

$$\dot{x}_3 = \frac{1}{L_a} V - \frac{R_a}{L_a} I_a - \frac{k_m}{L_a} \dot{\theta} \qquad (4)$$

$$= \frac{1}{L_a} V - \frac{R_a}{L_a} x_3 - \frac{k_m}{L_a} x_2$$

From (2) the output equation is:

$$\theta = x_1 \qquad (5)$$

The state equations can be obtained by expressing equations (4) and (5) in matrix form as:

$$\begin{bmatrix}\dot{x}_1\\ \dot{x}_2\\ \dot{x}_3\end{bmatrix} = \begin{bmatrix}0 & 1 & 0\\ 0 & -\frac{f}{J} & \frac{k_m}{J}\\ 0 & -\frac{k_m}{L_a} & -\frac{R_a}{L_a}\end{bmatrix}\begin{bmatrix}x_1\\ x_2\\ x_3\end{bmatrix} + \begin{bmatrix}0\\ 0\\ \frac{1}{L_a}\end{bmatrix}V$$

$$\theta = \begin{bmatrix}1 & 0 & 0\end{bmatrix}\begin{bmatrix}x_1\\ x_2\\ x_3\end{bmatrix}$$

Example 4

For a given complex system differential equation is

$$\frac{d^3x}{dt^3} + 3\frac{d^2x}{dt^2} + 5\frac{dx}{dt} + 7x(t) = u_1(t) + 3u_2(t) + 4u_3(t)$$

And the outputs are

$$y_1 = 2\frac{dx}{dt} + 3u_1(t) \quad \text{and} \quad y_2 = \frac{d^2x}{dt^2} + 6u_2(t) + u_3(t)$$

Represent the system in its state space form.

Solution

The differential equation can be rewritten as:

$$\dddot{x} = u_1(t) + 3u_2(t) + 4u_3(t) - 3\ddot{x} - 5\dot{x} - 7x \qquad (1)$$

The state variables can be chosen as:

$$x_1 = x$$
$$x_2 = \dot{x} \qquad (2)$$
$$x_3 = \ddot{x}$$

The derivatives of the state variables can be obtained as:

$$\dot{x}_1 = \dot{x}$$
$$\dot{x}_2 = \ddot{x} \qquad (3)$$
$$\dot{x}_3 = \dddot{x}$$

Substituting (1) and (2) in (3) and rearrange gives:

$$\dot{x}_1 = x_2$$

$$\dot{x}_2 = x_3 \tag{4}$$

$$\dot{x}_3 = u_1(t) + 3u_2(t) + 4u_3(t) - 3\ddot{x} - 5\dot{x} - 7x$$

$$= -7x_1 - 5x_2 - 3x_3 + u_1(t) + 3u_2(t) + 4u_3(t)$$

Substituting (2) in the output equations gives:

$$y_1 = 2x_2 + 3u_1(t)$$

$$y_2 = x_3 + 6u_2(t) + u_3(t) \tag{5}$$

The state equations can be obtained by expressing equations (4) and (5) in matrix form as:

$$\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \\ \dot{x}_3 \end{bmatrix} = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ -7 & -5 & -3 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} + \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 1 & 3 & 4 \end{bmatrix} \begin{bmatrix} u_1 \\ u_2 \\ u_3 \end{bmatrix}$$

$$\begin{bmatrix} y_1 \\ y_2 \end{bmatrix} = \begin{bmatrix} 0 & 2 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} + \begin{bmatrix} 3 & 0 & 0 \\ 0 & 6 & 1 \end{bmatrix} \begin{bmatrix} u_1 \\ u_2 \\ u_3 \end{bmatrix}$$

State Space representation of nth order systems of linear differential equation involving derivatives of forcing function

Consider a system whose differential equation involves derivatives of the forcing (input) function such as:

$$\frac{d^n y}{dt^n} + a_1 \frac{d^{n-1} y}{dt^{n-1}} + \ldots \ldots + a_n y = b_0 \frac{d^n u}{dt^n} + b_1 \frac{d^{n-1} u}{dt^{n-1}} + \ldots \ldots + b_{n-1} \frac{du}{dt} + b_n u \tag{1}$$

The state variables are chosen as:

$$\begin{aligned} x_1 &= y - \beta_0 u \\ x_2 &= \dot{y} - \beta_0 \dot{u} - \beta_1 u \\ x_3 &= \ddot{y} - \beta_0 \ddot{u} - \beta_1 \dot{u} - \beta_2 u \\ &\vdots \\ &\vdots \\ x_n &= y^{(n-1)} - \beta_0 u^{(n-1)} - \beta_1 u^{(n-2)} - \ldots \ldots - \beta_{n-2} \dot{u} - \beta_{n-1} u \end{aligned} \tag{2}$$

Where

$$
\begin{aligned}
&\beta_0 = b_0 \\
&\beta_1 = b_1 - a_1\beta_0 \\
&\beta_2 = b_2 - a_1\beta_1 - a_2\beta_0 \\
&\beta_3 = b_3 - a_1\beta_2 - a_2\beta_1 - a_3\beta_0 \\
&\vdots \\
&\beta_n = b_n - a_1\beta_{n-1} - \ldots - a_{n-1}\beta_1 - a_n\beta_0
\end{aligned}
\tag{3}
$$

Example 5

Consider the control system whose closed loop transfer function is

$$\frac{Y(s)}{U(s)} = \frac{s+3}{s^3 + 9s^2 + 24s + 20}$$

Obtain a state space representation of the system.

Solution

The corresponding differential equation can be obtained as:

$$\dddot{y} + 9\ddot{y} + 24\dot{y} + 20y = \dot{u} + 3u$$

From the differential equation:

$a_1 = 9$, $a_2 = 24$, $a_3 = 20$, $b_0 = 0$, $b_1 = 0$, $b_2 = 1$, $b_3 = 3$

From which:

$$
\begin{aligned}
&\beta_0 = b_0 = 0 \\
&\beta_1 = b_1 - a_1\beta_0 = 0 - 0 = 0 \\
&\beta_2 = b_2 - a_1\beta_1 - a_2\beta_0 = 1 - 0 - 0 = 1
\end{aligned}
$$

The differential equation can be rewritten as:

$$\dddot{y} = \dot{u} + 3u - 9\ddot{y} - 24\dot{y} - 20y \tag{1}$$

The state variables can be chosen as:

$$
\begin{aligned}
&x_1 = y - \beta_0 u = y \\
&x_2 = \dot{y} - \beta_0\dot{u} - \beta_1 u = \dot{y} \\
&x_3 = \ddot{y} - \beta_0\ddot{u} - \beta_1\dot{u} - \beta_2 u = \ddot{y} - u
\end{aligned}
\tag{2}
$$

The derivatives of the state variables can be obtained as:

$$\dot{x}_1 = \dot{y}$$

$$\dot{x}_2 = \ddot{y} \tag{3}$$

$$\dot{x}_3 = \dddot{y} - \dot{u}$$

From (2)

$$\begin{aligned} y &= x_1 \\ \dot{y} &= x_2 \\ \ddot{y} &= x_3 + u \end{aligned} \tag{4}$$

Substituting (4) and (1) in (3) and rearrange gives:

$$\dot{x}_1 = x_2$$

$$\dot{x}_2 = x_3 + u \tag{5}$$

$$\begin{aligned} \dot{x}_3 &= \dot{u} + 3u - 9\ddot{y} - 24\dot{y} - 20y - \dot{u} \\ &= -20x_1 - 24x_2 - 9(x_3 + u) + 3u \\ &= -20x_1 - 24x_2 - 9x_3 - 6u \end{aligned}$$

From (4) the output equation is:

$$y = x_1 \tag{6}$$

The state equations can be obtained by expressing equations (5) and (6) in matrix form as:

$$\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \\ \dot{x}_3 \end{bmatrix} = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ -20 & -24 & -9 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} + \begin{bmatrix} 0 \\ 1 \\ -6 \end{bmatrix} u$$

$$y = \begin{bmatrix} 1 & 0 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$$

Example 6

A system is represented by the following differential equation:

$$\dddot{y} + 5\ddot{y} + 4\dot{y} + y = \dddot{u} + 3\ddot{u} + \dot{u} + 2u$$

Obtain a state space representation of the system.

Solution

From the differential equation:

$a_1 = 5$, $a_2 = 4$, $a_3 = 1$, $b_0 = 1$, $b_1 = 3$, $b_2 = 1$, $b_3 = 2$

From which:

$$\beta_0 = b_0 = 1$$
$$\beta_1 = b_1 - a_1\beta_0 = 3 - 5(1) = 3 - 5 = -2$$
$$\beta_2 = b_2 - a_1\beta_1 - a_2\beta_0 = 1 - 5(-2) - 4(1) = 1 + 10 - 4 = 7$$

The differential equation can be rewritten as:

$$\dddot{y} = \dddot{u} + 3\ddot{u} + \dot{u} + 2u - 5\ddot{y} - 4\dot{y} - y \tag{1}$$

The state variables can be chosen as:

$$\begin{aligned} x_1 &= y - \beta_0 u = y - u \\ x_2 &= \dot{y} - \beta_0\dot{u} - \beta_1 u = \dot{y} - \dot{u} + 2u \\ x_3 &= \ddot{y} - \beta_0\ddot{u} - \beta_1\dot{u} - \beta_2 u = \ddot{y} - \ddot{u} + 2\dot{u} - 7u \end{aligned} \tag{2}$$

The derivatives of the state variables can be obtained as:

$$\begin{aligned} \dot{x}_1 &= \dot{y} - \dot{u} \\ \dot{x}_2 &= \ddot{y} - \ddot{u} + 2\dot{u} \\ \dot{x}_3 &= \dddot{y} - \dddot{u} + 2\ddot{u} - 7\dot{u} \end{aligned} \tag{3}$$

From (2)

$$\begin{aligned} y &= x_1 + u \\ \dot{y} &= x_2 + \dot{u} - 2u \\ \ddot{y} &= x_3 + \ddot{u} - 2\dot{u} + 7u \end{aligned} \tag{4}$$

Substituting (4) and (1) in (3) and rearrange gives:

$$\begin{aligned} \dot{x}_1 &= x_2 + \dot{u} - 2u - \dot{u} \\ &= x_2 - 2u \\ \dot{x}_2 &= x_3 + \ddot{u} - 2\dot{u} + 7u - \ddot{u} + 2\dot{u} \\ &= x_3 + 7u \\ \dot{x}_3 &= \dddot{u} + 3\ddot{u} + \dot{u} + 2u - 5\ddot{y} - 4\dot{y} - y - \dddot{u} + 2\ddot{u} - 7\dot{u} \\ &= 5\ddot{u} - 6\dot{u} + 2u - 5\ddot{y} - 4\dot{y} - y \\ &= 5\ddot{u} - 6\dot{u} + 2u - 5(x_3 + \ddot{u} - 2\dot{u} + 7u) - 4(x_2 + \dot{u} - 2u) - (x_1 + u) \end{aligned} \tag{5}$$

$$= 5\ddot{u} - 6\dot{u} + 2u - 5x_3 - 5\ddot{u} + 10\dot{u} - 35u - 4x_2 - 4\dot{u} + 8u - x_1 - u$$

$$= -x_1 - 4x_2 - 5x_3 - 26u$$

From (4) the output equation is:

$$y = x_1 + u \qquad (6)$$

The state equations can be obtained by expressing equations (5) and (6) in matrix form as:

$$\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \\ \dot{x}_3 \end{bmatrix} = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ -1 & -4 & -5 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} + \begin{bmatrix} -2 \\ 7 \\ -26 \end{bmatrix} u$$

$$y = \begin{bmatrix} 1 & 0 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} + u$$

Transfer function from state space equations

Consider the state space model

$$\dot{x} = Ax + Bu \qquad (1)$$

$$y = Cx + Du \qquad (2)$$

The transfer function can be obtained as follows:

Taking the Laplace transform of (1) assuming zero initial condition gives:

$$sX(s) = AX(s) + BU(s) \qquad (3)$$

$$Y(s) = CX(s) + DU(s) \qquad (4)$$

From (3)

$$sX(s) - AX(s) = BU(s)$$

$$(sI - A)X(s) = BU(s)$$

$$X(s) = (sI - A)^{-1}\, BU(s) \qquad (5)$$

Where is I identity matrix

Substituting (5) in (4) gives:

$$Y(s) = C(sI - A)^{-1}\, BU(s) + DU(s)$$

$$Y(s) = [C(sI - A)^{-1}\, B + D]U(s) \qquad (6)$$

The transfer function is obtained as:

$$G(s) = \frac{Y(s)}{U(s)} = C(sI - A)^{-1}\,B + D$$

$$= \frac{C[\boldsymbol{adj}(sI - A)]B}{|sI - A|} + D$$

$$= \frac{C[\boldsymbol{adj}(sI - A)]B + D|sI - A|}{|sI - A|}$$

Example 1

Determine the transfer function relating the output $y(t)$ to the input $u(t)$ for the system described by the state space equation below.

$$\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \end{bmatrix} = \begin{bmatrix} 2 & -1 \\ 3 & -4 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} + \begin{bmatrix} 1 \\ -1 \end{bmatrix} u$$

$$y = \begin{bmatrix} 1 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$$

Solution

From the state space equations:

$$A = \begin{bmatrix} 2 & -1 \\ 3 & -4 \end{bmatrix}, B = \begin{bmatrix} 1 \\ -1 \end{bmatrix}, C = \begin{bmatrix} 1 & 0 \end{bmatrix}, D = 0, I = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

The transfer function can be obtained using the relation:

$$G(s) = \frac{Y(s)}{U(s)} = \frac{C[\boldsymbol{adj}(sI - A)]B + D|sI - A|}{|sI - A|}$$

$$(sI - A) = s\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} - \begin{bmatrix} 2 & -1 \\ 3 & -4 \end{bmatrix} = \begin{bmatrix} s & 0 \\ 0 & s \end{bmatrix} - \begin{bmatrix} 2 & -1 \\ 3 & -4 \end{bmatrix} = \begin{bmatrix} (s-2) & 1 \\ -3 & s+4 \end{bmatrix}$$

$$adj(sI - A) = \begin{bmatrix} s+4 & -1 \\ 3 & s-2 \end{bmatrix}$$

$$|sI - A| = \begin{vmatrix} (s-2) & 1 \\ -3 & s+4 \end{vmatrix} = (s-2)(s+4) - 3(-1)$$

$$= s^2 + 4s - 2s - 8 + 3 = s^2 + 2s - 5$$

Therefore the transfer function can be obtained as:

$$G(s) = \frac{Y(s)}{U(s)} = \frac{\begin{bmatrix} 1 & 0 \end{bmatrix}\begin{bmatrix} s+4 & -1 \\ 3 & s-2 \end{bmatrix}\begin{bmatrix} 1 \\ -1 \end{bmatrix}}{s^2+2s-5}$$

$$= \frac{\begin{bmatrix} s+4 & -1 \end{bmatrix}\begin{bmatrix} 1 \\ -1 \end{bmatrix}}{s^2+2s-5}$$

$$= \frac{(s+4)(1)-1(-1)}{s^2+2s-5}$$

$$= \frac{s+5}{s^2+2s-5}$$

Example 1 can be solved using the following Matlab code:

```
A=[2 -1;3 -4]
B=[1;-1]
C=[1 0]
D=0
[num,den]=ss2tf(A,B,C,D)
G=tf(num,den)
```

Example 2

Determine the transfer function for the system described by the following state space equation:

$$\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \end{bmatrix} = \begin{bmatrix} -3 & 1 \\ -2 & 0 \end{bmatrix}\begin{bmatrix} x_1 \\ x_2 \end{bmatrix} + \begin{bmatrix} 0 \\ 1 \end{bmatrix} u$$

$$y = \begin{bmatrix} 1 & 0 \end{bmatrix}\begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$$

Solution

From the state space equations:

$$A = \begin{bmatrix} -3 & -1 \\ -2 & 0 \end{bmatrix}, B = \begin{bmatrix} 0 \\ 1 \end{bmatrix}, C = \begin{bmatrix} 1 & 0 \end{bmatrix}, D = 0, I = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

The transfer function can be obtained using the relation:

$$G(s) = \frac{Y(s)}{U(s)} = \frac{C[\boldsymbol{adj}(sI-A)]B+D|sI-A|}{|sI-A|}$$

$$(sI-A) = s\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} - \begin{bmatrix} -3 & -1 \\ -2 & 0 \end{bmatrix} = \begin{bmatrix} s & 0 \\ 0 & s \end{bmatrix} - \begin{bmatrix} -3 & -1 \\ -2 & 0 \end{bmatrix} = \begin{bmatrix} (s+3) & -1 \\ 2 & s \end{bmatrix}$$

$$adj(sI-A) = \begin{bmatrix} s & 1 \\ -2 & s+3 \end{bmatrix}$$

$$|sI - A| = \begin{vmatrix} (s+3) & -1 \\ 2 & s \end{vmatrix} = s(s+3) - 2(-1) = s^2 + 3s + 2$$

Therefore the transfer function can be obtained as:

$$G(s) = \frac{Y(s)}{U(s)} = \frac{\begin{bmatrix} 1 & 0 \end{bmatrix} \begin{bmatrix} s & 1 \\ -2 & s+3 \end{bmatrix} \begin{bmatrix} 0 \\ 1 \end{bmatrix}}{s^2+3s+2}$$

$$= \frac{\begin{bmatrix} s & 1 \end{bmatrix} \begin{bmatrix} 0 \\ 1 \end{bmatrix}}{s^2+3s+2}$$

$$= \frac{1}{s^2+3s+2}$$

Example 2 can be solved using the following Matlab code:

```
A=[-3 1;-2 0]
B=[0;1]
C=[1 0]
D=0
[num,den]=ss2tf(A,B,C,D)
G=tf(num,den)
```

Example 3

Determine the transfer function of a linear system described by the following state equations:

$$\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \\ \dot{x}_3 \end{bmatrix} = \begin{bmatrix} -2 & 0 & 1 \\ 1 & -2 & 0 \\ 1 & 1 & -1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} + \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix} u$$

$$y = \begin{bmatrix} 2 & 1 & -1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} + u$$

Solution

From the state equations given

$$A = \begin{bmatrix} -2 & 0 & 1 \\ 1 & -2 & 0 \\ 1 & 1 & -1 \end{bmatrix}, \qquad B = \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}, \qquad C = \begin{bmatrix} 2 & 1 & -1 \end{bmatrix}, \qquad D = 1$$

$$(sI - A) = \begin{bmatrix} s & 0 & 0 \\ 0 & s & 0 \\ 0 & 0 & s \end{bmatrix} - \begin{bmatrix} -2 & 0 & 1 \\ 1 & -2 & 0 \\ 1 & 1 & -1 \end{bmatrix} = \begin{bmatrix} s+2 & 0 & -1 \\ -1 & s+2 & 0 \\ -1 & -1 & s+1 \end{bmatrix}$$

$$adj(sI-A)=\begin{bmatrix} \begin{vmatrix} s+2 & 0 \\ -1 & s+1 \end{vmatrix} & -\begin{vmatrix} -1 & 0 \\ -1 & s+1 \end{vmatrix} & \begin{vmatrix} -1 & s+2 \\ -1 & -1 \end{vmatrix} \\ -\begin{vmatrix} 0 & -1 \\ -1 & s+1 \end{vmatrix} & \begin{vmatrix} s+2 & -1 \\ -1 & s+1 \end{vmatrix} & -\begin{vmatrix} s+2 & 0 \\ -1 & -1 \end{vmatrix} \\ \begin{vmatrix} 0 & -1 \\ s+2 & 0 \end{vmatrix} & -\begin{vmatrix} s+2 & -1 \\ -1 & 0 \end{vmatrix} & \begin{vmatrix} s+2 & 0 \\ -1 & s+2 \end{vmatrix} \end{bmatrix}^T$$

$$=\begin{bmatrix} s^2+3s+2 & s+1 & s+3 \\ 1 & s^2+3s+1 & s+2 \\ s+2 & 1 & s^2+4s+4 \end{bmatrix}^T$$

$$=\begin{bmatrix} s^2+3s+2 & 1 & s+2 \\ s+1 & s^2+3s+1 & 1 \\ s+3 & s+2 & s^2+4s+4 \end{bmatrix}$$

$$|sI-A|=\begin{vmatrix} s+2 & 0 & -1 \\ -1 & s+2 & 0 \\ -1 & -1 & s+1 \end{vmatrix}$$

$$=(s+2)\begin{vmatrix} s+2 & 0 \\ -1 & s+1 \end{vmatrix}-(0)\begin{vmatrix} -1 & 0 \\ -1 & s+1 \end{vmatrix}+(-1)\begin{vmatrix} -1 & s+2 \\ -1 & -1 \end{vmatrix}$$

$$=(s+2)(s^2+3s+2)-0-(s+3)$$

$$=s^3+3s^2+2s+2s^2+6s+4-s-3$$

$$=s^3+5s^2+7s+1$$

The transfer function can be obtained using the relation:

$$G(s)=\frac{Y(s)}{U(s)}=\frac{C[\boldsymbol{adj}(sI-A)]B+D|sI-A|}{|sI-A|}$$

$$=\frac{[2 \quad 1 \quad -1]\begin{bmatrix} s^2+3s+2 & 1 & s+2 \\ s+1 & s^2+3s+1 & 1 \\ s+3 & s+2 & s^2+4s+4 \end{bmatrix}\begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}+s^3+5s^2+7s+1}{s^3+5s^2+7s+1}$$

$$=\frac{[(2s^2+6s+4)+(s+1)-(s+3) \quad 2+(s^2+3s+1)-(s+2) \quad (2s+4)+1-(s^2+4s+4)]\begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}+s^3+5s^2+7s+1}{s^3+5s^2+7s+1}$$

$$=\frac{[2s^2+6s+2 \quad s^2+2s+1 \quad -s^2-2s+1]\begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}+s^3+5s^2+7s+1}{s^3+5s^2+7s+1}$$

$$=\frac{(2s^2+6s+2)+0+(-s^2-2s+1)+s^3+5s^2+7s+1}{s^3+5s^2+7s+1}$$

$$= \frac{s^2 + 4s + 3 + s^3 + 5s^2 + 7s + 1}{s^3 + 5s^2 + 7s + 1}$$

$$\frac{s^3 + 6s^2 + 11s + 4}{s^3 + 5s^2 + 7s + 1}$$

Example 3 can be solved using the following Matlab code:

```
A=[-2 0 1;1 -2 0;1 1 -1]
B=[1;0;1]
C=[2 1 -1]
D=1
[num,den]=ss2tf(A,B,C,D)
G=tf(num,den)
```

Example 4

Find the transfer function of the system represented by the following state and output equations:

$$\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \\ \dot{x}_3 \end{bmatrix} = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & -4 & -6 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} + \begin{bmatrix} 1 \\ -1 \\ 5 \end{bmatrix} u$$

$$y = \begin{bmatrix} 1 & 0 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$$

Solution

From the state equations given

$$A = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & -4 & -6 \end{bmatrix}, \qquad B = \begin{bmatrix} 1 \\ -1 \\ 5 \end{bmatrix}, \qquad C = \begin{bmatrix} 1 & 0 & 0 \end{bmatrix}, \qquad D = 0$$

$$(sI - A) = \begin{bmatrix} s & 0 & 0 \\ 0 & s & 0 \\ 0 & 0 & s \end{bmatrix} - \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & -4 & -6 \end{bmatrix} = \begin{bmatrix} s & -1 & 0 \\ 0 & s & -1 \\ 0 & 4 & s+6 \end{bmatrix}$$

$$adj(sI - A) = \begin{bmatrix} \begin{vmatrix} s & -1 \\ 4 & s+6 \end{vmatrix} & -\begin{vmatrix} 0 & -1 \\ 0 & s+6 \end{vmatrix} & \begin{vmatrix} 0 & s \\ 0 & 4 \end{vmatrix} \\ -\begin{vmatrix} -1 & 0 \\ 4 & s+6 \end{vmatrix} & \begin{vmatrix} s & 0 \\ 0 & s+6 \end{vmatrix} & -\begin{vmatrix} s & -1 \\ 0 & 4 \end{vmatrix} \\ \begin{vmatrix} -1 & 0 \\ s & -1 \end{vmatrix} & -\begin{vmatrix} s & 0 \\ 0 & -1 \end{vmatrix} & \begin{vmatrix} s & -1 \\ 0 & s \end{vmatrix} \end{bmatrix}^T$$

$$=\begin{bmatrix} s^2+6s+4 & 0 & 0 \\ s+6 & s^2+6s & -4s \\ 1 & s & s \end{bmatrix}^T$$

$$=\begin{bmatrix} s^2+6s+4 & s+6 & 1 \\ 0 & s^2+6s & s \\ 0 & -4s & s \end{bmatrix}$$

$$|sI-A| = \begin{vmatrix} s & -1 & 0 \\ 0 & s & -1 \\ 0 & 4 & s+6 \end{vmatrix}$$

$$= (s)\begin{vmatrix} s & -1 \\ 4 & s+6 \end{vmatrix} - (-1)\begin{vmatrix} 0 & -1 \\ 0 & s+6 \end{vmatrix} + (0)\begin{vmatrix} 0 & s \\ 0 & 4 \end{vmatrix}$$

$$= s(s^2+6s+4) - 0 - 0$$

$$= s^3+6s^2+4s$$

The transfer function can be obtained using the relation:

$$G(s) = \frac{Y(s)}{U(s)} = \frac{C[\boldsymbol{adj}(sI-A)]B + D|sI-A|}{|sI-A|}$$

$$= \frac{\begin{bmatrix} 1 & 0 & 0 \end{bmatrix}\begin{bmatrix} s^2+6s+4 & s+6 & 1 \\ 0 & s^2+6s & s \\ 0 & -4s & s \end{bmatrix}\begin{bmatrix} 1 \\ -1 \\ 5 \end{bmatrix}}{s^3+6s^2+4s}$$

$$= \frac{\begin{bmatrix} s^2+6s+4 & s+6 & 1 \end{bmatrix}\begin{bmatrix} 1 \\ -1 \\ 5 \end{bmatrix}}{s^3+6s^2+4s}$$

$$= \frac{(s^2+6s+4) - (s+6) + 5}{s^3+6s^2+4s}$$

$$= \frac{s^2+5s+3}{s^3+6s^2+4s}$$

Example 4 can be solved using the following Matlab code:

```
A=[ 0 1 0;0 0 1;0 -4 -6]
B=[1;-1;5]
C=[1 0 0]
D=0
[num,den]=ss2tf(A,B,C,D)
G=tf(num,den)
```

Characteristic equation

Consider the transfer function:

$$G(s) = \frac{Y(s)}{U(s)} = \frac{C[\boldsymbol{adj}(sI - A)]B + D|sI - A|}{|sI - A|}$$

The characteristic equation is given by:

$$|sI - A| = 0$$

Eigenvalues and Eigenvectors

Eigenvalues

The eigenvalues of an nxn system matrix A are the roots of the characteristic equation:

$$|\lambda I - A| = 0$$

Eigenvalues of system matrix (A) are actually poles of the system's transfer function. For a stable system all the eigenvalues must have negative real part.

Example 1

A system has it's a matrix given by

$$A = \begin{bmatrix} 4 & -1 & -1 \\ -1 & 2 & 1 \\ -1 & 1 & 2 \end{bmatrix}$$

Determine:

i. The characteristics equation
ii. Eigenvalues of A
iii. Stability of the system

Solution

i. $|\lambda I - A| =$

$$\left\| \begin{bmatrix} \lambda & 0 & 0 \\ 0 & \lambda & 0 \\ 0 & 0 & \lambda \end{bmatrix} - \begin{bmatrix} 4 & -1 & -1 \\ -1 & 2 & 1 \\ -1 & 1 & 2 \end{bmatrix} \right\| = \left\| \begin{bmatrix} \lambda - 4 & 1 & 1 \\ 1 & \lambda - 2 & -1 \\ 1 & -1 & \lambda - 2 \end{bmatrix} \right\| = 0$$

$$(\lambda-4)\begin{vmatrix}\lambda-2 & -1\\ -1 & \lambda-2\end{vmatrix}-(1)\begin{vmatrix}1 & -1\\ 1 & \lambda-2\end{vmatrix}+(1)\begin{vmatrix}1 & \lambda-2\\ 1 & -1\end{vmatrix}=0$$

$$\big((\lambda-4)((\lambda-2)(\lambda-2)-1)\big)-\big((\lambda-2)+1\big)+(-1-(\lambda-2))=0$$

$$\big((\lambda-4)((\lambda^2-4\lambda+4)-1)\big)-(\lambda-1)-(\lambda-1)=0$$

$$((\lambda^3-4\lambda^2+3\lambda)-4\lambda^2+16\lambda-12)-(2\lambda-2)=0$$

$$\lambda^3-8\lambda^2+17\lambda-10=0$$

ii. Factorizing the characteristics equation gives:

$$(\lambda-1)(\lambda-2)(\lambda-5)=0$$

Therefore the eigenvalues are: $\lambda=1,\ \lambda=2$ and $\lambda=5$

iii. Since the real parts of the eigenvalues are positive the system is unstable.

The eigenvalues in example 1 can be solved using the following Matlab code:

```
A=[ 4 -1 -1;-1 2 1;-1 1 2]
E=eig(A)
```

Eigenvectors

For each eigenvalue λ_i, there is a corresponding eigenvector $\underline{x_i}$ associated with it. The eigenvector is a set of solutions in the form of a non-zero vector. Eigenvalues are unique but eigenvectors are non-unique.

To find eigenvector the following relation is used:

$$Ax=\lambda x$$

$$(\lambda I-A)x=0$$

Example 2

A system matrix A given by

$$A=\begin{bmatrix}2 & 1 & 1\\ 2 & 3 & 4\\ -1 & -1 & -2\end{bmatrix}$$

Determine:

i. Characteristics equation
ii. Eigenvalues
iii. Eigenvectors

Solution

i. $|\lambda I - A| =$

$$\left|\begin{bmatrix}\lambda & 0 & 0\\ 0 & \lambda & 0\\ 0 & 0 & \lambda\end{bmatrix} - \begin{bmatrix}2 & 1 & 1\\ 2 & 3 & 4\\ -1 & -1 & -2\end{bmatrix}\right| = \left|\begin{bmatrix}\lambda-2 & -1 & -1\\ -2 & \lambda-3 & -4\\ 1 & 1 & \lambda+2\end{bmatrix}\right| = 0$$

$$(\lambda-2)\begin{vmatrix}\lambda-3 & -4\\ 1 & \lambda+2\end{vmatrix} - (-1)\begin{vmatrix}-2 & -4\\ 1 & \lambda+2\end{vmatrix} + (-1)\begin{vmatrix}-2 & \lambda-3\\ 1 & 1\end{vmatrix} = 0$$

$$\big((\lambda-2)((\lambda-3)(\lambda+2)+4)\big) + (-(2\lambda+4)+4) - (-2-(\lambda-3)) = 0$$

$$(\lambda-2)(\lambda^2-\lambda-6+4) - 2\lambda + \lambda - 1 = 0$$

$$(\lambda-2)(\lambda^2-\lambda-2) - \lambda - 1 = 0$$

$$\big((\lambda^3-\lambda^2-2\lambda) - 2\lambda^2 + 2\lambda + 4\big) - \lambda - 1 = 0$$

$$\lambda^3 - 3\lambda^2 - \lambda + 3 = 0$$

ii. Factorizing the characteristics equation gives:

$$(\lambda-1)(\lambda+1)(\lambda+3) = 0$$

Therefore the eigenvalues are: $\lambda = 1$, $\lambda = -1$ and $\lambda = 3$

iii. Eigenvectors: Using $(\lambda I - A)x = 0$

For $\lambda = 1$

$$(I - A)x = 0$$

$$\left[\begin{bmatrix}1 & 0 & 0\\ 0 & 1 & 0\\ 0 & 0 & 1\end{bmatrix} - \begin{bmatrix}2 & 1 & 1\\ 2 & 3 & 4\\ -1 & -1 & -2\end{bmatrix}\right]\begin{bmatrix}x_a\\ x_b\\ x_c\end{bmatrix} = \begin{bmatrix}0\\ 0\\ 0\end{bmatrix}$$

$$\begin{bmatrix}-1 & -1 & -1\\ -2 & -2 & -4\\ 1 & 1 & 3\end{bmatrix}\begin{bmatrix}x_a\\ x_b\\ x_c\end{bmatrix} = \begin{bmatrix}0\\ 0\\ 0\end{bmatrix}$$

$-x_a - x_b - x_c = 0 \qquad (1)$
$-2x_a - 2x_b - 4x_c = 0 \qquad (2)$
$x_a + x_b + 3x_c = 0 \qquad (3)$

Adding equation (1) and (3) gives: $2x_c = 0$

Therefore $x_c = 0$

Substituting $x_c = 0$ in (2) gives:

$$x_b = -x_a$$

Let

$$x_a = 1$$

Then

$$x_b = -1$$

Therefore the first eigenvector is:

$$v_1 = \begin{bmatrix} 1 \\ -1 \\ 0 \end{bmatrix}$$

For $\lambda = -1$

$$(-I - A)x = 0$$

$$\left[\begin{bmatrix} -1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & -1 \end{bmatrix} - \begin{bmatrix} 2 & 1 & 1 \\ 2 & 3 & 4 \\ -1 & -1 & -2 \end{bmatrix}\right]\begin{bmatrix} x_a \\ x_b \\ x_c \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

$$\begin{bmatrix} -3 & -1 & -1 \\ -2 & -4 & -4 \\ 1 & 1 & 1 \end{bmatrix}\begin{bmatrix} x_a \\ x_b \\ x_c \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

$-3x_a - x_b - x_c = 0 \qquad (4)$
$-2x_a - 4x_b - 4x_c = 0 \qquad (5)$
$x_a + x_b + x_c = 0 \qquad (6)$

Adding equation (4) and (6) gives: $-2x_a = 0$

Therefore $x_a = 0$

Substituting $x_a = 0$ in (2) gives:

$$x_c = -x_b$$

Let

$$x_b = 1$$

Then

$$x_c = -1$$

Therefore the second eigenvector is:

$$v_2 = \begin{bmatrix} 0 \\ 1 \\ -1 \end{bmatrix}$$

For $\lambda = 3$

$$(3I - A)x = 0$$

$$\left[\begin{bmatrix} 3 & 0 & 0 \\ 0 & 3 & 0 \\ 0 & 0 & 3 \end{bmatrix} - \begin{bmatrix} 2 & 1 & 1 \\ 2 & 3 & 4 \\ -1 & -1 & -2 \end{bmatrix}\right]\begin{bmatrix} x_a \\ x_b \\ x_c \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

$$\begin{bmatrix} 1 & -1 & -1 \\ -2 & 0 & -4 \\ 1 & 1 & 5 \end{bmatrix}\begin{bmatrix} x_a \\ x_b \\ x_c \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

$x_a - x_b - x_c = 0$ (7)

$-2x_a - 4x_c = 0$ (8)

$x_a + x_b + 5x_c = 0$ (9)

Adding equation (7) and (9) gives:

$$2x_a + 4x_c = 0$$

Then

$$x_a = -2x_c$$

Let

$$x_c = -1$$

Then

$$x_a = 2$$

From (7)

$x_b = x_a - x_c$ (10)

Substituting $x_c = -1$ and $x_a = -2$ in (10) gives

$$x_b = 3$$

Therefore the third eigenvector is:

$$v_3 = \begin{bmatrix} 2 \\ 3 \\ -1 \end{bmatrix}$$

The eigenvector of A is

$$V = [v_1 \quad v_2 \quad v_3] = \begin{bmatrix} 1 & 0 & 2 \\ -1 & 1 & 3 \\ 0 & -1 & -1 \end{bmatrix}$$

The eigenvalues of example 2 can be obtained using the following Matlab code:

```
A=[ 2 1 1;2 3 4;-1 -1 -2]
[V,D]=eig(A)
```

The above code produces a diagonal matrix *D* of eigenvalues and a full matrix *V* whose columns are the corresponding eigenvectors so that X*V = V*D.

State Equations Transformation

Consider the state space model

$$\dot{x} = Ax + Bu \tag{1}$$

$$y = Cx + Du \tag{2}$$

It is possible to transform the state-space representation with state vector $\boldsymbol{x}$ into a state-space representation with a state vector $\boldsymbol{z}$.

Suppose the relation between the state vector is defined by the following transformation:

$$x = Pz \tag{3}$$

Where P is the transformation matrix.

Substituting (3) into equations (1) and (2) gives:

$$P\dot{z} = APz + Bu$$

$$\dot{z} = P^{-1}APz + P^{-1}Bu \tag{4}$$

$$y = CPz + Du \tag{5}$$

Hence, the transformation yields a new space representation. If the newly formed A matrix is denoted by $A_z = P^{-1}AP$, then A and A_z have the following properties:

1. The same eigenvalues
2. The same determinant
3. The same trace (i.e. sum of elements on the principal diagonal).

Example 1

Given the system represented in state space as:

$$\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \\ \dot{x}_3 \end{bmatrix} = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ -2 & -5 & -7 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} + \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} u$$

$$y = \begin{bmatrix} 1 & 0 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$$

Transform the system to a new set of state variables $\boldsymbol{z}$, where the new variables are related to the original variables $\boldsymbol{x}$ as follows:

$$\left.\begin{aligned} z_1 &= 2x_1 \\ z_2 &= 3x_1 + 2x_2 \\ z_3 &= x_1 + 4x_2 + 5x_3 \end{aligned}\right\} \qquad \text{(a)}$$

Solution

$$A = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ -2 & -5 & -7 \end{bmatrix}, \quad B = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}, \quad C = \begin{bmatrix} 1 & 0 & 0 \end{bmatrix} \quad D = 0$$

From (a)

$$\begin{bmatrix} \dot{z}_1 \\ \dot{z}_2 \\ \dot{z}_3 \end{bmatrix} = \begin{bmatrix} 2 & 0 & 0 \\ 3 & 2 & 0 \\ 1 & 4 & 5 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} \qquad \text{(b)}$$

From (3)

$$z = P^{-1}x \qquad \text{(c)}$$

Comparing (b) with (c) gives:

$$P^{-1} = \begin{bmatrix} 2 & 0 & 0 \\ 3 & 2 & 0 \\ 1 & 4 & 5 \end{bmatrix}$$

$$P = (P^{-1})^{-1} = \frac{adjP}{|P|} = \frac{\begin{bmatrix} +\begin{vmatrix} 2 & 0 \\ 4 & 5 \end{vmatrix} & -\begin{vmatrix} 3 & 0 \\ 1 & 5 \end{vmatrix} & +\begin{vmatrix} 3 & 2 \\ 1 & 4 \end{vmatrix} \\ -\begin{vmatrix} 0 & 0 \\ 4 & 5 \end{vmatrix} & +\begin{vmatrix} 2 & 0 \\ 1 & 5 \end{vmatrix} & -\begin{vmatrix} 2 & 0 \\ 1 & 4 \end{vmatrix} \\ +\begin{vmatrix} 0 & 0 \\ 2 & 0 \end{vmatrix} & -\begin{vmatrix} 2 & 0 \\ 3 & 0 \end{vmatrix} & +\begin{vmatrix} 2 & 0 \\ 3 & 2 \end{vmatrix} \end{bmatrix}^T}{\begin{vmatrix} 2 & 0 & 0 \\ 3 & 2 & 0 \\ 1 & 4 & 5 \end{vmatrix}} = \frac{\begin{bmatrix} 10 & -15 & 10 \\ 0 & 10 & -8 \\ 0 & 0 & 4 \end{bmatrix}^T}{2\begin{vmatrix} 2 & 0 \\ 4 & 5 \end{vmatrix} - 0\begin{vmatrix} 3 & 0 \\ 1 & 5 \end{vmatrix} + 0\begin{vmatrix} 3 & 2 \\ 1 & 4 \end{vmatrix}}$$

$$= \frac{\begin{bmatrix} 10 & 0 & 0 \\ -15 & 10 & 0 \\ 10 & -8 & 4 \end{bmatrix}}{2(10) - 0 + 0} = \begin{bmatrix} 0.5 & 0 & 0 \\ -0.75 & 0.5 & 0 \\ 0.5 & -0.4 & 0.2 \end{bmatrix}$$

$$P^{-1}AP = \begin{bmatrix} 2 & 0 & 0 \\ 3 & 2 & 0 \\ 1 & 4 & 5 \end{bmatrix} \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ -2 & -5 & -7 \end{bmatrix} \begin{bmatrix} 0.5 & 0 & 0 \\ -0.75 & 0.5 & 0 \\ 0.5 & -0.4 & 0.2 \end{bmatrix}$$

$$= \begin{bmatrix} 0 & 2 & 0 \\ 0 & 3 & 2 \\ -10 & -24 & -31 \end{bmatrix} \begin{bmatrix} 0.5 & 0 & 0 \\ -0.75 & 0.5 & 0 \\ 0.5 & -0.4 & 0.2 \end{bmatrix} = \begin{bmatrix} -1.5 & 1 & 0 \\ -1.25 & 0.7 & 0.4 \\ -2.5 & 0.4 & -6.2 \end{bmatrix}$$

$$P^{-1}B = \begin{bmatrix} 2 & 0 & 0 \\ 3 & 2 & 0 \\ 1 & 4 & 5 \end{bmatrix} \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 5 \end{bmatrix}$$

$$CP = \begin{bmatrix} 1 & 0 & 0 \end{bmatrix} \begin{bmatrix} 0.5 & 0 & 0 \\ -0.75 & 0.5 & 0 \\ 0.5 & -0.4 & 0.2 \end{bmatrix} = \begin{bmatrix} 0.5 & 0 & 0 \end{bmatrix}$$

Hence

$$\dot{z} = \begin{bmatrix} -1.5 & 1 & 0 \\ -1.25 & 0.7 & 0.4 \\ -2.5 & 0.4 & -6.2 \end{bmatrix} z + \begin{bmatrix} 0 \\ 0 \\ 5 \end{bmatrix} u$$

$$y = \begin{bmatrix} 0.5 & 0 & 0 \end{bmatrix} z$$

State Space in diagonal form

A system in state space can be represented in diagonal form so that all system state are separated or uncoupled as illustrated in the example below:

$$\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \\ \dot{x}_3 \end{bmatrix} = \begin{bmatrix} -4 & 0 & 0 \\ 0 & -3 & 0 \\ 0 & 0 & -2 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} + \begin{bmatrix} 0 \\ 0 \\ 24 \end{bmatrix} r \quad \Rightarrow \quad \begin{aligned} \dot{x}_1 &= -4x_1 \\ \dot{x}_2 &= -3x_2 \\ \dot{x}_3 &= -2x_3 + 24r \end{aligned}$$

From the example, it can be seen that every state equation is a function of each state variable. Therefore each differential equation can be solve independently

Consider the state space model

$$\dot{x} = Ax + Bu \tag{1}$$

$$y = Cx + Du \tag{2}$$

The state space model in (1) and (2) can be transform into diagonal form as follows:

$$\dot{z} = P^{-1}APz + P^{-1}Bu \tag{4}$$

$$y = CPz + Du \tag{5}$$

In general, there are several methods of finding the transformation matrix P. It can be formed by the use of the eigenvectors of A, that is

$$P = V$$

Note that the matrix $P^{-1}AP$ is a diagonal matrix whose diagonal elements are the eigenvalues of A.

Example 1

Consider a system represented in the state space by the following equations:

$$\dot{x} = \begin{bmatrix} -3 & 1 \\ 1 & -3 \end{bmatrix} x + \begin{bmatrix} 1 \\ 2 \end{bmatrix} u$$

$$y = \begin{bmatrix} 2 & 3 \end{bmatrix} x$$

Represent the system in diagonal form.

Solution

$A = \begin{bmatrix} -3 & 1 \\ 1 & -3 \end{bmatrix}$, $B = \begin{bmatrix} 1 \\ 2 \end{bmatrix}$, $\quad C = \begin{bmatrix} 2 & 3 \end{bmatrix}$

Eigenvalues

$$|\lambda I - A| = 0$$

$$|\lambda I - A| = \left| \begin{bmatrix} \lambda & 0 \\ 0 & \lambda \end{bmatrix} - \begin{bmatrix} -3 & 1 \\ 1 & -3 \end{bmatrix} \right| = \begin{vmatrix} \lambda + 3 & -1 \\ -1 & \lambda + 3 \end{vmatrix} = 0$$
$$(\lambda + 3)(\lambda + 3) - 1 = 0$$
$$\lambda^2 + 6\lambda + 9 - 1 = 0$$
$$\lambda^2 + 6\lambda + 8 = 0$$
$$(\lambda + 2)(\lambda + 4) = 0$$

Hence $\lambda_1 = -2$ and $\lambda_2 = -4$

Eigenvectors

$$(\lambda I - A)x = 0$$

For $\lambda = -2$

$$\left[\begin{bmatrix} -2 & 0 \\ 0 & -2 \end{bmatrix} - \begin{bmatrix} -3 & 1 \\ 1 & -3 \end{bmatrix} \right] \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 1 & -1 \\ -1 & 1 \end{bmatrix} \begin{bmatrix} x_a \\ x_b \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

$$x_a - x_b = 0 \qquad (1)$$

$$-x_a + x_b = 0 \qquad (2)$$

From (1) or (2)

$$x_b = x_a$$

Let

$$x_a = 1$$

Then

$$x_b = 1$$

Therefore the first eigenvector is:

$$v_1 = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

For $\lambda = -4$

$$\left[\begin{bmatrix} -4 & 0 \\ 0 & -4 \end{bmatrix} - \begin{bmatrix} -3 & 1 \\ 1 & -3 \end{bmatrix}\right]\begin{bmatrix} x_a \\ x_b \end{bmatrix} = \begin{bmatrix} -1 & -1 \\ -1 & -1 \end{bmatrix}\begin{bmatrix} x_a \\ x_b \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

$$-x_a - x_b = 0 \qquad (1)$$
$$-x_a - x_b = 0 \qquad (2)$$

From (1) or (2)

$$x_b = -x_a$$

Let

$$x_a = 1$$

Then

$$x_b = -1$$

Therefore the second eigenvector is:

$$v_1 = \begin{bmatrix} 1 \\ -1 \end{bmatrix}$$

The eigenvector of A is therefore:

$$V = [v_1 \quad v_2] = \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}$$

Transformation matrix

$$P = V = \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}$$

$$P^{-1} = \frac{adjP}{|P|} = \frac{\begin{bmatrix} -1 & -1 \\ -1 & 1 \end{bmatrix}}{\begin{vmatrix} 1 & 1 \\ 1 & -1 \end{vmatrix}} = \frac{\begin{bmatrix} -1 & -1 \\ -1 & 1 \end{bmatrix}}{-2} = \begin{bmatrix} 0.5 & 0.5 \\ 0.5 & -0.5 \end{bmatrix}$$

$$P^{-1}AP = \begin{bmatrix} 0.5 & 0.5 \\ 0.5 & -0.5 \end{bmatrix} \begin{bmatrix} -3 & 1 \\ 1 & -3 \end{bmatrix} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix} = \begin{bmatrix} -1 & -1 \\ -2 & 2 \end{bmatrix} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix} = \begin{bmatrix} -2 & 0 \\ 0 & -4 \end{bmatrix}$$

$$P^{-1}B = \begin{bmatrix} 0.5 & 0.5 \\ 0.5 & -0.5 \end{bmatrix} \begin{bmatrix} 1 \\ 2 \end{bmatrix} = \begin{bmatrix} 1.5 \\ -0.5 \end{bmatrix}$$

$$CP = [2 \quad 3] \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix} = [5 \quad -1]$$

Hence

$$\dot{z} = \begin{bmatrix} -2 & 0 \\ 0 & -4 \end{bmatrix} z + \begin{bmatrix} 1.5 \\ -0.5 \end{bmatrix} u$$

$$y = [5 \quad -1] z$$

Example 1 can be solved using the following Matlab code:

```
A=[-3 1;1 -3]
B=[1;2]
C=[2 3]
D=0
G=ss(A,B,C,D)
canon(G,'modal')
```

Example 2

Diagonalize the following system

$$\dot{x} = \begin{bmatrix} -5 & -5 & 4 \\ 2 & 0 & -2 \\ 0 & -2 & -1 \end{bmatrix} x + \begin{bmatrix} -1 \\ 2 \\ -2 \end{bmatrix} r$$

$$y = [-1 \quad 1 \quad 2] x$$

Solution

$$A = \begin{bmatrix} -5 & -5 & 4 \\ 2 & 0 & -2 \\ 0 & -2 & -1 \end{bmatrix},\ B = \begin{bmatrix} -1 \\ 2 \\ -2 \end{bmatrix},\quad C = \begin{bmatrix} -1 & 1 & 2 \end{bmatrix}$$

Eigenvalues

$$|\lambda I - A| = 0$$

$$|\lambda I - A| = \left| \begin{bmatrix} \lambda & 0 & 0 \\ 0 & \lambda & 0 \\ 0 & 0 & \lambda \end{bmatrix} - \begin{bmatrix} -5 & -5 & 4 \\ 2 & 0 & -2 \\ 0 & -2 & -1 \end{bmatrix} \right| = \begin{vmatrix} \lambda + 5 & 5 & -4 \\ -2 & \lambda & 2 \\ 0 & 2 & \lambda + 1 \end{vmatrix} = 0$$

$$(\lambda + 5) \begin{vmatrix} \lambda & 2 \\ 2 & \lambda + 1 \end{vmatrix} - 5 \begin{vmatrix} -2 & 2 \\ 0 & \lambda + 1 \end{vmatrix} + -4 \begin{vmatrix} -2 & \lambda \\ 0 & 2 \end{vmatrix} = 0$$

$$(\lambda + 5)(\lambda^2 + \lambda - 4) - 5(-2\lambda - 2) - 4(-4) = 0$$

$$(\lambda^3 + \lambda^2 - 4\lambda + 5\lambda^2 + 5\lambda - 20) + (10\lambda + 10) + 16 = 0$$

$$\lambda^3 + 6\lambda^2 + 11\lambda + 6 = 0$$

$$(\lambda + 1)(\lambda + 2)(\lambda + 3) = 0$$

Hence $\lambda_1 = -1$, $\lambda_2 = -2$ and $\lambda_3 = -3$

Eigenvectors

$$(\lambda I - A)x = 0$$

For $\lambda = -1$

$$\left[\begin{bmatrix} -1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & -1 \end{bmatrix} - \begin{bmatrix} -5 & -5 & 4 \\ 2 & 0 & -2 \\ 0 & -2 & -1 \end{bmatrix} \right] \begin{bmatrix} x_a \\ x_b \\ x_c \end{bmatrix} = \begin{bmatrix} 4 & 5 & -4 \\ -2 & -1 & 2 \\ 0 & 2 & 0 \end{bmatrix} \begin{bmatrix} x_a \\ x_b \\ x_c \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

$$4x_a + 5x_b - 4x_c = 0 \qquad (1)$$
$$-2x_a - x_b + 2x_c = 0 \qquad (2)$$
$$2x_b = 0 \qquad (3)$$

From (3)

$$x_b = 0 \qquad (4)$$

Substituting (4) in (1) and (2) gives:

$$4x_a - 4x_c = 0 \qquad (5)$$
$$-2x_a + 2x_c = 0 \qquad (6)$$

From (5) or (6)

$$x_a = x_c$$

Let

$$x_a = 1$$

Then

$$x_c = 1$$

Therefore the first eigenvector is:

$$v_1 = \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}$$

For $\lambda = -2$

$$\left[\begin{bmatrix} -2 & 0 & 0 \\ 0 & -2 & 0 \\ 0 & 0 & -2 \end{bmatrix} - \begin{bmatrix} -5 & -5 & 4 \\ 2 & 0 & -2 \\ 0 & -2 & -1 \end{bmatrix}\right] \begin{bmatrix} x_a \\ x_b \\ x_c \end{bmatrix} = \begin{bmatrix} 3 & 5 & -4 \\ -2 & -2 & 2 \\ 0 & 2 & -1 \end{bmatrix} \begin{bmatrix} x_a \\ x_b \\ x_c \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

$$3x_a + 5x_b - 4x_c = 0 \qquad (1)$$
$$-2x_a - 2x_b + 2x_c = 0 \qquad (2)$$
$$2x_b - x_c = 0 \qquad (3)$$

From (3)

$$x_c = 2x_b \qquad (4)$$

Let

$$x_b = 1 \qquad (5)$$

Then

$$x_c = 2 \qquad (6)$$

Substituting (5) and (6) in (1) or (2) gives:

$$x_a = 1 \qquad (7)$$

Therefore the second eigenvector is:

$$v_2 = \begin{bmatrix} 1 \\ 1 \\ 2 \end{bmatrix}$$

For $\lambda = -3$

$$\left[\begin{bmatrix}-3 & 0 & 0\\ 0 & -3 & 0\\ 0 & 0 & -3\end{bmatrix} - \begin{bmatrix}-5 & -5 & 4\\ 2 & 0 & -2\\ 0 & -2 & -1\end{bmatrix}\right]\begin{bmatrix}x_a\\ x_b\\ x_c\end{bmatrix} = \begin{bmatrix}2 & 5 & -4\\ -2 & -3 & 2\\ 0 & 2 & -2\end{bmatrix}\begin{bmatrix}x_a\\ x_b\\ x_c\end{bmatrix} = \begin{bmatrix}0\\ 0\\ 0\end{bmatrix}$$

$$2x_a + 5x_b - 4x_c = 0 \qquad (1)$$
$$-2x_a - 3x_b + 2x_c = 0 \qquad (2)$$
$$2x_b - 2x_c = 0 \qquad (3)$$

From (3)

$$x_c = x_b \qquad (4)$$

Let

$$x_b = 1 \qquad (5)$$

Then

$$x_c = 1 \qquad (6)$$

Substituting (5) and (6) in (1) or (2) gives:

$$x_a = -0.5 \qquad (7)$$

Therefore the third eigenvector is:

$$v_3 = \begin{bmatrix}-0.5\\ 1\\ 1\end{bmatrix}$$

The eigenvector matrix of A is therefore:

$$V = [v_1 \quad v_2 \quad v_3] = \begin{bmatrix}1 & 1 & -0.5\\ 0 & 1 & 1\\ 1 & 2 & 1\end{bmatrix}$$

Transformation matrix

$$P = V = \begin{bmatrix}1 & 1 & -0.5\\ 0 & 1 & 1\\ 1 & 2 & 1\end{bmatrix}$$

$$P^{-1} = \frac{adjP}{|P|} = \frac{\begin{bmatrix} \begin{vmatrix} 1 & 1 \\ 2 & 1 \end{vmatrix} & -\begin{vmatrix} 0 & 1 \\ 1 & 1 \end{vmatrix} & \begin{vmatrix} 0 & 1 \\ 1 & 2 \end{vmatrix} \\ -\begin{vmatrix} 1 & -0.5 \\ 2 & 1 \end{vmatrix} & \begin{vmatrix} 1 & -0.5 \\ 1 & 1 \end{vmatrix} & -\begin{vmatrix} 1 & 1 \\ 1 & 2 \end{vmatrix} \\ \begin{vmatrix} 1 & -0.5 \\ 1 & 1 \end{vmatrix} & -\begin{vmatrix} 1 & -0.5 \\ 0 & 1 \end{vmatrix} & \begin{vmatrix} 1 & 1 \\ 0 & 1 \end{vmatrix} \end{bmatrix}^T}{\begin{vmatrix} 1 & 1 & -0.5 \\ 0 & 1 & 1 \\ 1 & 2 & 1 \end{vmatrix}} = \frac{\begin{bmatrix} -1 & 1 & -1 \\ -2 & 1.5 & -1 \\ 1.5 & -1 & 1 \end{bmatrix}^T}{\begin{vmatrix} 1 & 1 \\ 2 & 1 \end{vmatrix} - \begin{vmatrix} 0 & 1 \\ 1 & 1 \end{vmatrix} - 0.5\begin{vmatrix} 0 & 1 \\ 1 & 2 \end{vmatrix}}$$

$$= \frac{\begin{bmatrix} -1 & 1 & -1 \\ -2 & 1.5 & -1 \\ 1.5 & -1 & 1 \end{bmatrix}^T}{-1+1+0.5} = \frac{\begin{bmatrix} -1 & -2 & 1.5 \\ 1 & 1.5 & -1 \\ -1 & -1 & 1 \end{bmatrix}}{0.5} = \begin{bmatrix} -2 & -4 & 3 \\ 2 & 3 & -2 \\ -2 & -2 & 2 \end{bmatrix}$$

$$P^{-1}AP = \begin{bmatrix} -2 & -4 & 3 \\ 2 & 3 & -2 \\ -2 & -2 & 2 \end{bmatrix}\begin{bmatrix} -5 & -5 & 4 \\ 2 & 0 & -2 \\ 0 & -2 & -1 \end{bmatrix}\begin{bmatrix} 1 & 1 & -0.5 \\ 0 & 1 & 1 \\ 1 & 2 & 1 \end{bmatrix} = \begin{bmatrix} 2 & 4 & -3 \\ -4 & -6 & 4 \\ 6 & 6 & -6 \end{bmatrix}\begin{bmatrix} 1 & 1 & -0.5 \\ 0 & 1 & 1 \\ 1 & 2 & 1 \end{bmatrix}$$
$$= \begin{bmatrix} -1 & 0 & 0 \\ 0 & -2 & 0 \\ 0 & 0 & -3 \end{bmatrix}$$

$$P^{-1}B = \begin{bmatrix} -2 & -4 & 3 \\ 2 & 3 & -2 \\ -2 & -2 & 2 \end{bmatrix}\begin{bmatrix} -1 \\ 2 \\ -2 \end{bmatrix} = \begin{bmatrix} -12 \\ 8 \\ -6 \end{bmatrix}$$

$$CP = \begin{bmatrix} -1 & 1 & 2 \end{bmatrix}\begin{bmatrix} 1 & 1 & -0.5 \\ 0 & 1 & 1 \\ 1 & 2 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 4 & 3.5 \end{bmatrix}$$

Hence

$$\dot{z} = \begin{bmatrix} -1 & 0 & 0 \\ 0 & -2 & 0 \\ 0 & 0 & -3 \end{bmatrix} z + \begin{bmatrix} -12 \\ 8 \\ -6 \end{bmatrix} u$$

$$y = \begin{bmatrix} 1 & 4 & 3.5 \end{bmatrix} z$$

Time Domain Solution of State Equations

The process of determining the time domain solution of the state equation is similar to that of solving a simple first order differential equation.

Consider a simple first order differential equation given by:

$$\dot{x} = ax \qquad (1)$$

Taking Laplace transform of (1) gives:

$$sX(s) - x(0) = aX(s)$$
$$sX(s) - aX(s) = x(0)$$

$$(s - a)X(s) = x(0)$$

$$X(s) = \frac{x(0)}{(s-a)} \qquad (2)$$

Taking the inverse Laplace transform of (2) gives the solution of the differential equation in (1) as:

$$x(t) = e^{at}x(0) \qquad (3)$$

Consider the state equation:

$$\dot{x}(t) = Ax(t) + Bu(t) \qquad (4)$$

Homogenous system

A homogenous system is a system without input that is unforced system.

For homogenous system $u(t) = 0$ hence the state equation in (1) becomes:

$$\dot{x}(t) = Ax(t) \qquad (5)$$

Similar to (3) the solution of (5) can be written as:

$$x(t) = e^{At}x(0)$$

$$= \Phi(t)x(0) \qquad (6)$$

$\Phi(t) = e^{At}$ is called the **state-transition matrix**, since it performs a transformation on $x(0)$, taking x from the initial state, $x(0)$, to the state $x(t)$ at any time, t.

Each term of $\Phi(t)$ is the sum of exponentials generated by the system's poles (eigenvalues).

$\Phi(t)$ has the following properties:

i. $\Phi(0) = I$
ii. $\dot{\Phi}(0) = A$

The **state-transition matrix** $\Phi(t)$ can be obtained using two techniques:

1. Classical technique
2. Laplace technique

Classical technique

The following example illustrates the classical technique:

Example 1

Obtain the state-transition matrix $\Phi(t)$ of the following system using classical method:

$$\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \end{bmatrix} = \begin{bmatrix} 0 & 1 \\ -2 & -3 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$$

Solution

$$A = \begin{bmatrix} 0 & 1 \\ -2 & -3 \end{bmatrix}$$

Eigenvalues

$$|\lambda I - A| = 0$$

$$|\lambda I - A| = \left| \begin{bmatrix} \lambda & 0 \\ 0 & \lambda \end{bmatrix} - \begin{bmatrix} 0 & 1 \\ -2 & -3 \end{bmatrix} \right| = \begin{vmatrix} \lambda & -1 \\ 2 & \lambda + 3 \end{vmatrix} = 0$$
$$\lambda(\lambda + 3) + 2 = 0$$
$$\lambda^2 + 3\lambda + 2 = 0$$
$$(\lambda + 1)(\lambda + 2) = 0$$

Hence $\lambda_1 = -1$ and $\lambda_2 = -2$

Since each term of the state-transition matrix is the sum of responses generated by the poles (eigenvalues), we assume a state-transition matrix of the:

$$\Phi(t) = \begin{bmatrix} k_1 e^{-t} + k_2 e^{-2t} & k_3 e^{-t} + k_4 e^{-2t} \\ k_5 e^{-t} + k_6 e^{-2t} & k_7 e^{-t} + k_8 e^{-2t} \end{bmatrix} \quad (1)$$

Using property (i) that is

$$\Phi(0) = I$$
$$\begin{bmatrix} k_1 + k_2 & k_3 + k_4 \\ k_5 + k_6 & k_7 + k_8 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

Hence

$$k_1 + k_2 = 1 \quad (2)$$

$$k_3 + k_4 = 0 \Rightarrow k_3 = -k_4 \quad (3)$$

$$k_5 + k_6 = 0 \Rightarrow k_5 = -k_6 \quad (4)$$

$$k_7 + k_8 = 1 \Rightarrow k_7 = 1 - k_8 \qquad (5)$$

Using property (ii) that is

$$\dot{\Phi}(0) = A$$

$$\begin{bmatrix} -k_1 e^{-t} - 2k_2 e^{-2t} & -k_3 e^{-t} - 2k_4 e^{-2t} \\ -k_5 e^{-t} - 2k_6 e^{-2t} & -k_7 e^{-t} - 2k_8 e^{-2t} \end{bmatrix} = \begin{bmatrix} 0 & 1 \\ -2 & -3 \end{bmatrix}$$

$$-k_1 - 2k_2 = 0 \ \Rightarrow k_1 = -2k_2 \qquad (6)$$

$$-k_3 - 2k_4 = 1 \qquad (7)$$

$$-k_5 - 2k_6 = -2 \qquad (8)$$

$$-k_7 - 2k_8 = -3 \qquad (9)$$

Adding (2) and (6) gives:

$$k_2 = -1$$

Hence

$$k_1 = 2$$

Adding (3) and (7) gives:

$$k_4 = -1$$

Hence

$$k_3 = 1$$

Adding (4) and (8) gives:

$$k_6 = 2$$

Hence

$$k_5 = -2$$

Adding (5) and (9) gives:

$$k_8 = 2$$

Hence

$$k_7 = -1$$

Therefore

$$\Phi(t) = \begin{bmatrix} 2e^{-t} - e^{-2t} & e^{-t} - e^{-2t} \\ -2e^{-t} + 2e^{-2t} & -e^{-t} + 2e^{-2t} \end{bmatrix}$$

Laplace technique

Consider the homogenous state equation:

$$\dot{x} = Ax \qquad (1)$$

Taking the Laplace transform of (1) gives:

$$sX(s) - x(0) = AX(s)$$

$$sX(s) - AX(s) = x(0)$$

$$(sI - A)X(s) = x(0)$$

$$X(s) = (sI - A)^{-1}x(0) \qquad (2)$$

$$X(s) = \Phi(s)x(0) \qquad (3)$$

Where

$$\Phi(s) = (sI - A)^{-1} \qquad (4)$$

Taking inverse Laplace transform gives:

$$\mathcal{L}^{-1}[X(s)] = \mathcal{L}^{-1}[(sI - A)^{-1}]x(0)$$

$$x(t) = \mathcal{L}^{-1}[(sI - A)^{-1}]x(0)$$

$$x(t) = \mathcal{L}^{-1}[\Phi(s)]x(0)$$

$$\Phi(t) = \mathcal{L}^{-1}[\Phi(s)]$$

Example 2

Obtain the state-transition matrix $\Phi(t)$ of the following system using Laplace method:

$$\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \end{bmatrix} = \begin{bmatrix} 0 & 1 \\ -2 & -3 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$$

Solution

$$A = \begin{bmatrix} 0 & 1 \\ -2 & -3 \end{bmatrix}$$

$$(sI - A) = \begin{bmatrix} s & 0 \\ 0 & s \end{bmatrix} - \begin{bmatrix} 0 & 1 \\ -2 & -3 \end{bmatrix} = \begin{bmatrix} s & -1 \\ 2 & s+3 \end{bmatrix}$$

$$(sI - A)^{-1} = \frac{adj(sI - A)}{|sI - A|} = \frac{\begin{bmatrix} s+3 & 1 \\ -2 & s \end{bmatrix}}{\begin{vmatrix} s & -1 \\ 2 & s+3 \end{vmatrix}} \frac{\begin{bmatrix} s+3 & 1 \\ -2 & s \end{bmatrix}}{s^2 + 3s + 2} = \begin{bmatrix} \frac{s+3}{s^2+3s+2} & \frac{1}{s^2+3s+2} \\ \frac{-2}{s^2+3s+2} & \frac{s}{s^2+3s+2} \end{bmatrix}$$

$$= \begin{bmatrix} \frac{s+3}{(s+1)(s+2)} & \frac{1}{(s+1)(s+2)} \\ \frac{-2}{(s+1)(s+2)} & \frac{s}{(s+1)(s+2)} \end{bmatrix}$$

Using partial fraction expansion:

$$\frac{s+3}{(s+1)(s+2)} = \frac{A}{(s+1)} + \frac{B}{(s+2)}$$

$$(s+2)A + (s+1)B = s+3$$

$$(A+B)s + (2A+B) = s+3$$

Comparing coefficients gives:

$$A + B = 1 \Rightarrow B = 1 - A \qquad (1)$$

$$2A + B = 3 \qquad (2)$$

Equation (2) – (1) gives:

$$A = 2$$

Hence

$$B = 1 - 2 = -1$$

$$\frac{1}{(s+1)(s+2)} = \frac{A}{(s+1)} + \frac{B}{(s+2)}$$

$$(s+2)A + (s+1)B = 1$$

$$(A+B)s + (2A+B) = 1$$

Comparing coefficients gives:

$$A + B = 0 \Rightarrow B = -A \tag{3}$$

$$2A + B = 1 \tag{4}$$

Equation (4) – (3) gives:

$$A = 1$$

Hence

$$B = -1$$

$$\frac{-2}{(s+1)(s+2)} = \frac{A}{(s+1)} + \frac{B}{(s+2)}$$

$$(s+2)A + (s+1)B = -2$$

$$(A+B)s + (2A+B) = -2$$

Comparing coefficients gives:

$$A + B = 0 \Rightarrow B = -A \tag{5}$$

$$2A + B = -2 \tag{6}$$

Equation (6) – (5) gives:

$$A = -2$$

Hence

$$B = 2$$

$$\frac{s}{(s+1)(s+2)} = \frac{A}{(s+1)} + \frac{B}{(s+2)}$$

$$(s+2)A + (s+1)B = s$$

$$(A+B)s + (2A+B) = s$$

Comparing coefficients gives:

$$A + B = 1 \Rightarrow B = 1 - A \tag{7}$$

$$2A + B = 0 \tag{8}$$

Equation (8) – (7) gives:

$$A = -1$$

Hence

$$B = 2$$

Therefore

$$(sI - A)^{-1} = \begin{bmatrix} \frac{2}{(s+1)} - \frac{1}{(s+2)} & \frac{1}{(s+1)} - \frac{1}{(s+2)} \\ -\frac{2}{(s+1)} + \frac{1}{(s+2)} & -\frac{1}{(s+1)} + \frac{2}{(s+2)} \end{bmatrix}$$

$$\Phi(t) = \mathcal{L}^{-1}[\Phi(s)] = \mathcal{L}^{-1}[(sI - A)^{-1}] = \begin{bmatrix} 2e^{-t} - e^{-2t} & e^{-t} - e^{-2t} \\ -2e^{-t} + 2e^{-2t} & -e^{-t} + 2e^{-2t} \end{bmatrix}$$

Non-homogenous system

A non-homogenous system is a system with input that is forced system.

The non-homogenous state equation is given by:

$$\dot{x}(t) = Ax(t) + Bu(t) \tag{1}$$

Equation (1) can be written as:

$$\dot{x}(t) - Ax(t) = Bu(t)$$

Multiplying both sides by e^{-At}gives:

$$e^{-At}[\dot{x}(t) - Ax(t)] = e^{-At}Bu(t)$$

$$\frac{d}{dt}[e^{-At}x(t)] = e^{-At}Bu(t)$$

Integrating both sides gives:

$$[e^{-At}x(t)]|_0^t = \int_0^t e^{-A\tau}Bu(\tau)$$

$$e^{-At}x(t) - x(0) = \int_0^t e^{-A\tau}Bu(\tau)$$

Multiplying both sides by e^{At}gives:

$$x(t) - e^{At}x(0) = e^{At}\int_0^t e^{-A\tau}Bu(\tau)$$

$$x(t) = e^{At}x(0) + \int_0^t e^{A(t-\tau)}Bu(\tau)$$

$$x(t) = \Phi(t)x(0) + \int_0^t \Phi(t-\tau)Bu(\tau) \qquad (2)$$

Equation (2) is the time domain solution of non-homogenous state equation:

Where

$$\Phi(t) = e^{At}$$

$$\Phi(t-\tau) = e^{A(t-\tau)}$$

Example 1

Consider the system below:

$$\begin{bmatrix} \dot{x}_1(t) \\ \dot{x}_2(t) \end{bmatrix} = \begin{bmatrix} 0 & 1 \\ -8 & -6 \end{bmatrix} \begin{bmatrix} x_1(t) \\ x_2(t) \end{bmatrix} + \begin{bmatrix} 0 \\ 1 \end{bmatrix} u(t)$$

$$y(t) = \begin{bmatrix} 1 & 0 \end{bmatrix} \begin{bmatrix} x_1(t) \\ x_2(t) \end{bmatrix}$$

$$x(0) = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$$

$$u(\tau) = 1$$

i. Determine the state transition matrices, $\Phi(t)$.
ii. Solve for $x(t)$ and $y(t)$.
iii. Use Matlab to plot the response $y(t)$

Solution

$$A = \begin{bmatrix} 0 & 1 \\ -8 & -6 \end{bmatrix}, \; B = \begin{bmatrix} 0 \\ 1 \end{bmatrix}, \quad C = \begin{bmatrix} 1 & 0 \end{bmatrix}, \; x(0) = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$$

$$(sI - A) = \begin{bmatrix} s & 0 \\ 0 & s \end{bmatrix} - \begin{bmatrix} 0 & 1 \\ -8 & -6 \end{bmatrix} = \begin{bmatrix} s & -1 \\ 8 & s+6 \end{bmatrix}$$

$$(sI - A)^{-1} = \frac{adj(sI - A)}{|sI - A|} = \frac{\begin{bmatrix} s+6 & 1 \\ -8 & s \end{bmatrix}}{\begin{vmatrix} s & -1 \\ 8 & s+6 \end{vmatrix}} \frac{\begin{bmatrix} s+6 & 1 \\ -8 & s \end{bmatrix}}{s^2 + 6s + 8} = \begin{bmatrix} \frac{s+6}{s^2+6s+8} & \frac{1}{s^2+6s+8} \\ \frac{-8}{s^2+6s+8} & \frac{s}{s^2+6s+8} \end{bmatrix}$$

$$= \begin{bmatrix} \frac{s+6}{(s+2)(s+4)} & \frac{1}{(s+2)(s+4)} \\ \frac{-8}{(s+2)(s+4)} & \frac{s}{(s+2)(s+4)} \end{bmatrix}$$

Using partial fraction expansion:

$$\frac{s+6}{(s+2)(s+4)} = \frac{A}{(s+2)} + \frac{B}{(s+4)}$$

$$(s+4)A + (s+2)B = s+6$$

$$(A+B)s + (4A+2B) = s+6$$

Comparing coefficients gives:

$$A + B = 1 \Rightarrow B = 1 - A \qquad (1)$$

$$4A + 2B = 6 \Rightarrow 2A + B = 3 \qquad (2)$$

Equation (2) – (1) gives:

$$A = 2$$

Hence

$$B = 1 - 2 = -1$$

$$\frac{1}{(s+2)(s+4)} = \frac{A}{(s+2)} + \frac{B}{(s+4)}$$

$$(s+4)A + (s+2)B = 1$$

$$(A+B)s + (4A+2B) = 1$$

Comparing coefficients gives:

$$A + B = 0 \Rightarrow B = -A \qquad (3)$$

$$4A + 2B = 1 \qquad (4)$$

Equation (4) – 2 x equation (3) gives:

$$A = 0.5$$

Hence

$$B = -0.5$$

$$\frac{-8}{(s+2)(s+4)} = \frac{A}{(s+2)} + \frac{B}{(s+4)}$$

$$(s+4)A + (s+2)B = -8$$

$$(A+B)s + (4A+2B) = -8$$

Comparing coefficients gives:

$$A + B = 0 \Rightarrow B = -A \qquad (5)$$

$$4A + 2B = -8 \Rightarrow 2A + B = -4 \qquad (6)$$

Equation (6) – (5) gives:

$$A = -4$$

Hence

$$B = 4$$

$$\frac{s}{(s+2)(s+4)} = \frac{A}{(s+2)} + \frac{B}{(s+4)}$$

$$(s+4)A + (s+2)B = s$$

$$(A+B)s + (4A+2B) = s$$

Comparing coefficients gives:

$$A + B = 1 \Rightarrow B = 1 - A \qquad (7)$$

$$4A + 2B = 0 \Rightarrow 2A + B = 0 \qquad (8)$$

Equation (8) – (7) gives:

$$A = -1$$

Hence

$$B = 2$$

Therefore

$$(sI - A)^{-1} = \begin{bmatrix} \frac{2}{(s+2)} - \frac{1}{(s+4)} & \frac{0.5}{(s+2)} - \frac{0.5}{(s+4)} \\ -\frac{4}{(s+2)} + \frac{4}{(s+4)} & -\frac{1}{(s+2)} + \frac{2}{(s+4)} \end{bmatrix}$$

$$\Phi(t) = \mathcal{L}^{-1}[\Phi(s)] = \mathcal{L}^{-1}[(sI - A)^{-1}] = \begin{bmatrix} 2e^{-2t} - e^{-4t} & 0.5e^{-2t} - 0.5e^{-4t} \\ -4e^{-2t} + 4e^{-4t} & -e^{-2t} + 2e^{-4t} \end{bmatrix}$$

$$x(t) = \Phi(t)x(0) + \int_0^t \Phi(t-\tau)Bu(\tau)\, d\tau$$

$$= \begin{bmatrix} 2e^{-2t} - e^{-4t} & 0.5e^{-2t} - 0.5e^{-4t} \\ -4e^{-2t} + 4e^{-4t} & -e^{-2t} + 2e^{-4t} \end{bmatrix} \begin{bmatrix} 1 \\ 0 \end{bmatrix} + \int_0^t \begin{bmatrix} 2e^{-2(t-\tau)} - e^{-4(t-\tau)} & 0.5e^{-2(t-\tau)} - 0.5e^{-4(t-\tau)} \\ -4e^{-2(t-\tau)} + 4e^{-4(t-\tau)} & -e^{-2(t-\tau)} + 2e^{-4(t-\tau)} \end{bmatrix} \begin{bmatrix} 0 \\ 1 \end{bmatrix}$$

$$= \begin{bmatrix} 2e^{-2t} - e^{-4t} \\ -4e^{-2t} + 4e^{-4t} \end{bmatrix} + \int_0^t \begin{bmatrix} 0.5e^{-2(t-\tau)} - 0.5e^{-4(t-\tau)} \\ -e^{-2(t-\tau)} + 2e^{-4(t-\tau)} \end{bmatrix} d\tau$$

$$= \begin{bmatrix} 2e^{-2t} - e^{-4t} \\ -4e^{-2t} + 4e^{-4t} \end{bmatrix} + \begin{bmatrix} 0.25e^{-2(t-\tau)} - 0.125e^{-4(t-\tau)} \\ -0.5e^{-2(t-\tau)} + 0.5e^{-4(t-\tau)} \end{bmatrix}\Bigg|_0^t$$

$$= \begin{bmatrix} 2e^{-2t} - e^{-4t} \\ -4e^{-2t} + 4e^{-4t} \end{bmatrix} + \left[\begin{bmatrix} 0.25e^{-2(t-t)} - 0.125e^{-4(t-t)} \\ -0.5e^{-2(t-t)} + 0.5e^{-4(t-t)} \end{bmatrix} - \begin{bmatrix} 0.25e^{-2(t-0)} - 0.125e^{-4(t-0)} \\ -0.5e^{-2(t-0)} + 0.5e^{-4(t-0)} \end{bmatrix}\right]$$

$$= \begin{bmatrix} 2e^{-2t} - e^{-4t} \\ -4e^{-2t} + 4e^{-4t} \end{bmatrix} + \left[\begin{bmatrix} 0.25 - 0.125 \\ -0.5 + 0.5 \end{bmatrix} - \begin{bmatrix} 0.25e^{-2t} - 0.125e^{-4t} \\ -0.5e^{-2t} + 0.5e^{-4t} \end{bmatrix}\right]$$

$$= \begin{bmatrix} 2e^{-2t} - e^{-4t} \\ -4e^{-2t} + 4e^{-4t} \end{bmatrix} + \begin{bmatrix} 0.125 - 0.25e^{-2t} + 0.125e^{-4t} \\ 0.5e^{-2t} - 0.5e^{-4t} \end{bmatrix}$$

$$= \begin{bmatrix} 0.125 + 1.75e^{-2t} - 0.875e^{-4t} \\ -3.5e^{-2t} + 3.5e^{-4t} \end{bmatrix}$$

$$y(t) = Cx_1(t)$$

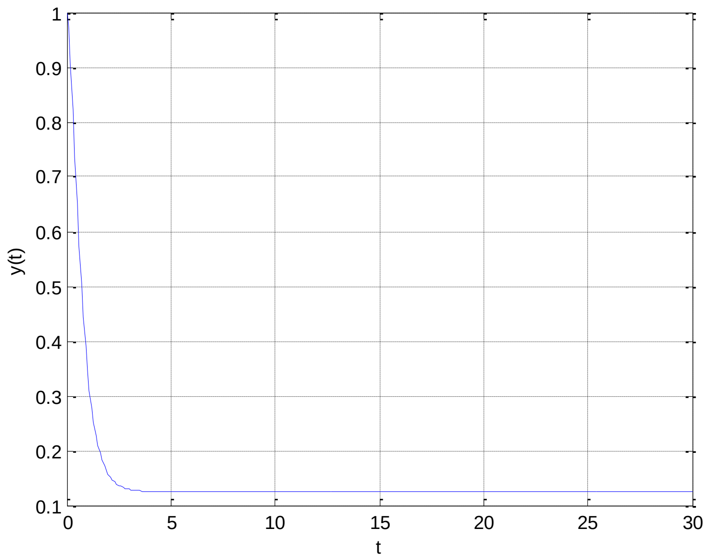
$$y(t) = \begin{bmatrix} 1 & 0 \end{bmatrix} \begin{bmatrix} 0.125 + 1.75e^{-2t} - 0.875e^{-4t} \\ -3.5e^{-2t} + 3.5e^{-4t} \end{bmatrix}$$

$$= 0.125 + 1.75e^{-2t} - 0.875e^{-4t}$$

The output can be plotted using the following Matlab code:

```
t=(0:0.1:30);
y=0.125+1.75*exp(-2*t)-0.875*exp(-4*t);
plot(t,y)
grid
xlabel('t')
ylabel('y(t)')
```

Running the code the following result will be obtained:



y(t)
1
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0
5
10
15
20
25
30
t